A LOCALLY NOETHERIAN SCHEME IS COHEN-MACAULAY EXACTLY WHEN ITS DUALIZING COMPLEX HAS A SINGLE NONZERO COHOMOLOGY SHEAF

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AWT (sources/stacks-duality/pinned-original-source.tex line 6214).

Allowed standard prerequisites (editorial).

- Localisation of a dualizing complex and the Cohen-Macaulay criterion for it. Dualizing Complexes, Lemmas labelled dualizing-lemma-dualizing-localize and dualizing-lemma-apply-CM, cited at proof lines 6230-6236 and 6264-6266. Give (2) and the Cohen-Macaulayness of the single cohomology sheaf in the connected case.
- Vanishing of a finite type module stalk spreading to a neighbourhood. Modules, Lemma labelled modules-lemma-finite-type-stalk-zero, cited at proof lines 6244-6246. Shrinks the affine neighbourhood so that only one cohomology sheaf survives.
- The definition of a Cohen-Macaulay module. Cohomology of Schemes, Definition labelled coherent-definition-Cohen-Macaulay, cited at proof lines 6266-6268. Identifies the resulting module as Cohen-Macaulay.

Source reference. Original lemma 0A7G: sources/stacks-chow/tags.txt, source lines 12879–12879. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6232 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AWS: sources/stacks-chow/tags.txt, source lines 13730–13730. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6235 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 01B9: sources/stacks-chow/tags.txt, source lines 1718–1718. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6247 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 0343: sources/stacks-chow/tags.txt, source lines 4078–4078. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6267 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a locally Noetherian scheme with dualizing complex $\omega_X^\bullet$.*

(1) *X is Cohen-Macaulay $\Leftrightarrow$ $\omega_X^\bullet$ locally has a unique nonzero cohomology sheaf,*
(2) *$\mathcal{O}_{X,x}$ is Cohen-Macaulay $\Leftrightarrow$ $\omega_{X,x}^\bullet$ has a unique nonzero cohomology,*
(3) *$U = \{x \in X \mid \mathcal{O}_{X,x}$ is Cohen-Macaulay$\}$ is open and Cohen-Macaulay.*

If X is connected and Cohen-Macaulay, then there is an integer n and a coherent Cohen-Macaulay $\mathcal{O}_X$-module ω_X such that $\omega_X^\bullet = \omega_X[-n]$.

Proof. By definition and Dualizing Complexes, Lemma 0A7G for every $x \in X$ the complex $\omega_{X,x}^\bullet$ is a dualizing complex over $\mathcal{O}_{X,x}$. By Dualizing Complexes, Lemma 0AWS we see that (2) holds.

To see (3) assume that $\mathcal{O}_{X,x}$ is Cohen-Macaulay. Let n_x be the unique integer such that $H^{n_x}(\omega_{X,x}^\bullet)$ is nonzero. For an affine neighbourhood $V \subset X$ of x we have $\omega_X^\bullet|_V$ is in $D^b_{Coh}(\mathcal{O}_V)$ hence there are finitely many nonzero coherent modules $H^i(\omega_X^\bullet)|_V$. Thus after shrinking V we may assume only H^{n_x} is nonzero, see Modules, Lemma 01B9. In this way we see that $\mathcal{O}_{X,v}$ is Cohen-Macaulay for every $v \in V$. This proves that U is open as well as a Cohen-Macaulay scheme.

Proof of (1). The implication $\Leftarrow$ follows from (2). The implication $\Rightarrow$ follows from the discussion in the previous paragraph, where we showed that if $\mathcal{O}_{X,x}$ is Cohen-Macaulay, then in a neighbourhood of x the complex $\omega_X^\bullet$ has only one nonzero cohomology sheaf.

Assume X is connected and Cohen-Macaulay. The above shows that the map $x \mapsto n_x$ is locally constant. Since X is connected it is constant, say equal to n. Setting $\omega_X = H^n(\omega_X^\bullet)$ we see that the lemma holds because ω_X is Cohen-Macaulay by Dualizing Complexes, Lemma 0AWS (and Cohomology of Schemes, Definition 0343). $\square$